

# Correlation and Prediction for Isobaric Vapor–Liquid Equilibria of the Diethyl Ether + Methanol + 1-Butanol Ternary System and the Constituent Binary Systems at 101.325 kPa

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**ABSTRACT:** The vapor–liquid equilibrium (VLE) data for the diethyl ether + methanol + 1-butanol ternary system and three constituent binary systems were measured at different liquid phase compositions using a dynamic recirculating still at 101.325 kPa. The activity coefficients of the solution were correlated with the Wilson, nonrandom two-liquid (NRTL), Margules, van Laar, and universal quasichemical activity coefficient (UNIQUAC) models through the fit of least-squares method. In addition, the VLE data of the ternary system were also predicted from these binary interaction parameters of Wilson, NRTL, Margules, van Laar, and UNIQUAC model parameters without any additional adjustment, which obtained the calculated vapor-phase compositions and bubble points compared with the measured values. The calculated bubble points with the model parameters of activity coefficients were in good agreement with the experimental data. The ASOG group contribution method also was used for prediction of the three binary systems. The thermodynamic consistency of the experimental VLE data was checked out by means of the Wisniak's *L*–*W* test for the binary systems and the Wisniak–Tamir's modification of McDermott–Ellis test for the ternary system, respectively.

## INTRODUCTION

The reaction of carbon monoxide reduction with hydrogen using the catalyst of metal copper or zinc is the most common and important technology for synthesis of methanol in chemical industrial process. However, the resultant products contain the byproduct diethyl ether and 1-butanol, therefore, the vapor–liquid equilibrium (VLE) data of diethyl ether + methanol + 1-butanol ternary system and the constituent binary systems are indispensable in distillation separation and design process to the product of carbon monoxide reduction. Many attempts have been made to investigate the phase behavior of the constituent binary systems. Donham et al. investigated that the *p*–*V*–*T*–*x* relations of the methanol–1-butanol and diethyl ether–1-butanol systems were determined at the liquid–vapor phase boundaries from near their atmospheric boiling points to the highest temperature and pressure at which the liquid and vapor coexist.<sup>1</sup> Onken et al. reported isothermal (303.15 K) and isobaric (93.325 kPa) VLE data measured for the binary system diethyl ether–methanol by using a recirculation still. The activity coefficients were fitted by using the Margules, van Laar, Wilson, NRTL, and UNIQUAC equations.<sup>2</sup> Smith et al. investigated the total pressure VLE data reported at 298, 338, and 388 K for the diethyl ether–methanol binary system. The Mixon–Gumowski–Carpenter and Barker methods were used to reduce the experimental *pT**x* data.<sup>3</sup> Tojo et al. explored density, refractive index, and speed of sound of the diethyl ether–methanol binary mixture from 288.15 to 298.15 K at atmospheric pressure measured over the whole composition range.<sup>4,5</sup> Arm et al. investigated the total vapor pressures, the heats of mixing, and the refractive indices of the system diethyl ether–methanol at 298.15 K.

The partial pressures, activity coefficients, excess free energies, entropy functions, and excess volumes were calculated.<sup>6</sup> Pettit explored the boiling points for the diethyl ether–methanol binary system at different liquid-phase compositions at 98.49 kPa.<sup>7</sup> Eckert et al. reported an asymmetric isothermal flow calorimeter was used to obtain excess enthalpies of binary liquid mixtures of diethyl ether, methanol, and 1-butanol in the dilute region, and these data were used to calculate partial molar excess enthalpies at infinite dilution.<sup>8</sup> Moreover, the limiting activity coefficients, heats of mixing, excess volumes, excess isentropic compressions, excess isobaric heat capacities, and excess molar enthalpies for methanol–1-butanol binary mixtures were measured at 101.325 kPa between 293.15 and 323.15 K.<sup>9–12</sup> In addition, Arce et al. explored that the isobaric VLE data were determined at 101.32 kPa and predicted for the ternary system 1-butanol + methanol + TAME (tert-amyl methyl ether) or MTBE (methyl tert-butyl ether) with 1-butanol as a possible entrainer for the separation of TAME or MTBE and methanol or ethanol by extractive, respectively. The VLE data were satisfactorily correlated using the Wilson, NRTL, and UNIQUAC equations for liquid phase activity coefficients and adequately predicted using the ASOG, UNIFAC, UNIFAC–Dortmund, and UNIFAC–Lyngby group contribution methods.<sup>13–16</sup> In our previous work, we investigated that the VLE data for the associating systems containing alkyl alcohol were correlated and predicted.<sup>17,18</sup>

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Herein, the VLE data for diethyl ether + methanol + 1-butanol system and constituent binary systems were measured by the total pressure–temperature–liquid and vapor phase composition ( $p$ ,  $T$ ,  $x$ ,  $y$ ) method using the recirculation still at 101.325 kPa. Hayden–O’Connell (HOC) model was used to correct the nonideality of vapor phase.<sup>19</sup> However, the nonideality of liquid phase was corrected by the calculation of its activity coefficient, which was obtained based on Wilson,<sup>20</sup> nonrandom two-liquid (NRTL),<sup>21</sup> Margules,<sup>22</sup> van Laar,<sup>23</sup> and universal quasichemical activity coefficient (UNIQUAC)<sup>24</sup> models as the function of  $T$ ,  $x$  through the nonlinear fit of the least-squares method, respectively. Wilson, NRTL, Margules, van Laar, and UNIQUAC models were applied to correlate the VLE data for the three constituent binary systems. And the VLE data of the ternary system were also predicted from these binary interaction parameters of Wilson, NRTL, Margules, van Laar, and UNIQUAC models without any additional adjustment, which obtained the calculated vapor-phase compositions and bubble points compared with the measured values. In addition, the ASOG<sup>25</sup> model also was used for prediction of the three binary systems. The excess Gibbs free energy of binary systems in the overall range of liquid-phase mole composition was calculated by the liquid activity coefficient correlation to the Wilson model parameters using the experimental data. The thermodynamic consistency of the experimental VLE data reported in this work was checked out by means of the Wisniak’s L–W test for the binary systems,<sup>26</sup> and the Wisniak–Tamir’s modification of McDermott–Ellis test for the ternary systems.<sup>27,28</sup>

## EXPERIMENTAL SECTION

**Materials.** Diethyl ether was supplied by Sigma-Aldrich with a nominal purity of more than  $\omega = 0.997$  (mass fraction). Both methanol and 1-butanol were provided by Sigma-Aldrich with a nominal purity of more than  $\omega = 0.998$  (mass fraction). All the chemicals were degassed using ultrasound for several hours and then dried on a molecular sieve (pore diameter 0.3 nm from Fluka) to remove all possible traces of moisture before use, but no other treatments were applied. The densities and refractive indices at 298.15 K and normal boiling points at 101.325 kPa of the pure component compared with the literature values of Riddick et al.<sup>29</sup> The measured values are approximately in agreement with those of the literature. The measurement method of the composition dependence of densities and refractive indices has previously been reported.<sup>17</sup>

**Apparatus and Procedure.** An all-glass VLE apparatus model 602, manufactured by Fischer (Germany), was used in the equilibrium determinations. The temperature was measured with a Thermolyzer S2541 (Frontek) temperature meter (resolution 0.005 K) equipped with a Pt-100 probe. The thermometers were calibrated at an accredited calibration laboratory (Quality and Technique Bureau, Anhui), with a calibration uncertainty of 0.015 K. The uncertainty in the temperature measurement of the system is estimated to be  $\pm 0.07$  K, due to the uncertainty of the calibration, the location of the probes, and the small pressure fluctuations. The Pt-100 probe was located at the bottom of the packed section of the equilibrium chamber. The pressure was measured with a Fischer digital manometer with accuracy of  $\pm 0.01$  kPa. The uncertainty of the pressure measurement was  $\pm 0.07$  kPa, according to the data provided by the manufacturer of the pressure measurement devices. The pressure

measurement system was calibrated with a DHPPC-2 pressure calibrator. Including the calibration uncertainty, the uncertainty in the pressure measurement system is  $\pm 0.15$  kPa. In this circulation apparatus, the solution was heated to its boiling point by a 250 W immersion heater. The vapor–liquid mixture flowed through an extended contact line that guarantees an intense phase exchange and then entered a separation chamber whose construction prevented an entrainment of liquid particles into the vapor phase. The separated gas and liquid phases were condensed and returned to a mixing chamber, where they were stirred by a magnetic stirrer, and returned again to the immersion heater. The distillation was carried out at 101.325 kPa. The experimental procedures were similar to those of earlier measurements.<sup>13</sup> The system VLE attained was kept at boiling point for 20 min to ensure the stationary state, and then we extracted samples of condensed vapor and liquid phase with syringes. The compositions of condensed vapor and liquid phases for the binary and ternary mixtures at equilibrium were analyzed with a HP 6850A gas chromatograph equipped with series connected flame ionization detectors (FID) and an autosampler. The GC column used was an HP-1 (cross-linked methyl siloxane, length 30 m, column inner diameter 0.25 mm, film thickness 1.0  $\mu\text{m}$ ). The injector and detectors were at 420 and 450 K, respectively. The oven was operated at variable programmed temperature, from 383 to 453 K at a rate of 10 K  $\cdot$  min<sup>−1</sup>. Helium (99.999% purity) was used as carrier gas with a flow of 45 mL  $\cdot$  min<sup>−1</sup>. The gravimetric calibration mixtures were prepared in 2-mL vials with approximately 1 mL of toluene as a solvent for the GC calibration for all systems measured. The accuracy of the analysis of the compositions of the phases was estimated to be  $\pm 0.001$  in mole fraction.

## RESULTS AND DISCUSSION

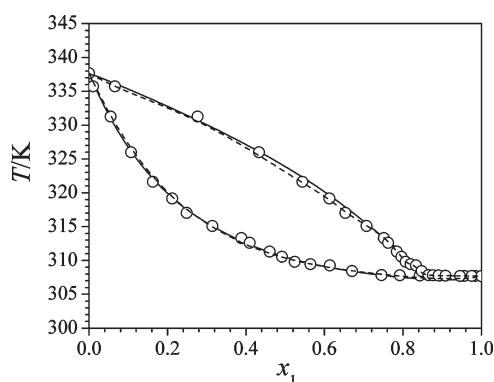
**Correlation and Prediction of VLE Data of the Binary Systems.** The activity coefficients  $\gamma_i$  of the components were calculated from

$$y_i \hat{\phi}_i^V p = x_i \gamma_i \phi_i^L p_i^s \exp \left[ -\frac{V_i^L (p - p_i^s)}{RT} \right] \quad (1)$$

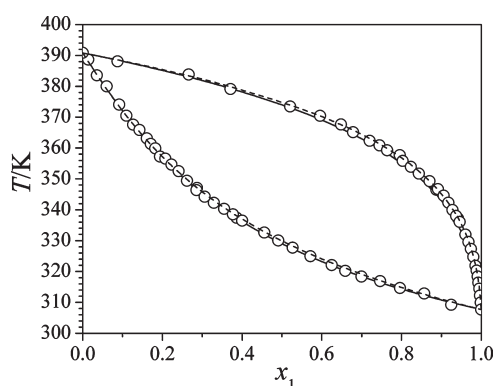
where  $x_i$  and  $y_i$  are the liquid- and vapor-phase mole fractions of component  $i$  in equilibrium,  $\hat{\phi}_i^V$  is the fugacity coefficient of component  $i$  in the vapor mixture,  $\phi_i^L$  is the fugacity coefficient of component  $i$  at saturation,  $V_i^L$  is the molar volume of component  $i$  in the liquid phase,  $R$  is the universal gas constant and  $T$  is the experimental temperature,  $p$  is the total pressure, and  $p_i^s$  is the vapor pressure of pure component  $i$ . These vapor pressures were calculated from the Antoine equation. The constants  $A_i$ ,  $B_i$ , and  $C_i$  were obtained from Reid et al.<sup>30</sup> The Poynting correction factor was also included in the calculation of fugacity at the reference state. The liquid molar volumes were evaluated from the modified Rackett equation.<sup>31</sup> To account for the nonideal behavior, Hayden–O’Connell (HOC) model<sup>19</sup> was used to correct the nonideality of vapor phase.<sup>32</sup> For the three binary systems, the activity coefficients were correlated with the Wilson,<sup>20</sup> NRTL,<sup>21</sup> Margules,<sup>22</sup> van Laar,<sup>23</sup> and UNIQUAC<sup>24</sup> equations. The optimum interaction parameters were obtained by minimization of the objective function (OF) by means of

**Table 1. VLE Data for the Diethyl Ether (1) + Methanol (2), Diethyl Ether (1) + 1-Butanol (2), and Methanol (1) + 1-Butanol (2) Three Binary Systems at 101.325 kPa: Measured Liquid-Phase Mole Fractions  $x_1$ , Measured Vapor-Phase Mole Fractions  $y_1$ , Experimental Boiling Point Temperature  $T_{\text{exp}}$ , and Experimental Activity Coefficients  $\gamma_1$  and  $\gamma_2$**

| $x_1$                             | $y_1$  | $T_{\text{exp}}$ | $\gamma_1$ | $\gamma_2$ | $x_1$  | $y_1$  | $T_{\text{exp}}$ | $\gamma_1$ | $\gamma_2$ |
|-----------------------------------|--------|------------------|------------|------------|--------|--------|------------------|------------|------------|
| diethyl ether (1) + methanol (2)  |        |                  |            |            |        |        |                  |            |            |
| 0.0000                            | 0.0000 | 337.66           |            | 1.0000     | 0.5243 | 0.8072 | 309.83           | 1.4206     | 1.2727     |
| 0.0105                            | 0.0656 | 335.71           | 2.3044     | 1.0001     | 0.5645 | 0.8196 | 309.45           | 1.3613     | 1.3412     |
| 0.0552                            | 0.2765 | 331.26           | 2.2346     | 1.0016     | 0.6139 | 0.8333 | 309.28           | 1.2931     | 1.4463     |
| 0.1070                            | 0.4332 | 326.00           | 2.1478     | 1.0062     | 0.6701 | 0.8475 | 308.45           | 1.2223     | 1.6029     |
| 0.1626                            | 0.5438 | 321.62           | 2.0498     | 1.0153     | 0.7454 | 0.8653 | 307.84           | 1.1402     | 1.9024     |
| 0.2118                            | 0.6127 | 319.15           | 1.9606     | 1.0277     | 0.7923 | 0.8768 | 307.82           | 1.0971     | 2.1660     |
| 0.2492                            | 0.6531 | 317.04           | 1.8923     | 1.0402     | 0.8433 | 0.8908 | 307.80           | 1.0579     | 2.5541     |
| 0.3144                            | 0.7072 | 315.10           | 1.7736     | 1.0700     | 0.8927 | 0.9083 | 307.77           | 1.0286     | 3.0783     |
| 0.3884                            | 0.7518 | 313.32           | 1.6425     | 1.1192     | 0.9563 | 0.9456 | 307.73           | 1.0051     | 4.0947     |
| 0.4097                            | 0.7623 | 312.63           | 1.6059     | 1.1370     | 0.9842 | 0.9752 | 307.73           | 1.0007     | 4.7220     |
| 0.4604                            | 0.7843 | 311.32           | 1.5212     | 1.1879     | 1.0000 | 1.0000 | 307.71           | 1.0000     |            |
| 0.4919                            | 0.7962 | 310.57           | 1.4707     | 1.2265     |        |        |                  |            |            |
| diethyl ether (1) + 1-butanol (2) |        |                  |            |            |        |        |                  |            |            |
| 0.0000                            | 0.0000 | 390.79           |            | 1.0000     | 0.3061 | 0.8902 | 344.24           | 0.9768     | 0.9946     |
| 0.0138                            | 0.1044 | 388.67           | 0.9382     | 0.9999     | 0.3290 | 0.9045 | 342.24           | 0.9786     | 0.9940     |
| 0.0354                            | 0.2456 | 383.53           | 0.9431     | 0.9999     | 0.3550 | 0.9182 | 340.30           | 0.9802     | 0.9930     |
| 0.0603                            | 0.3800 | 380.00           | 0.9481     | 0.9998     | 0.3864 | 0.9319 | 337.37           | 0.9824     | 0.9924     |
| 0.0915                            | 0.5135 | 374.07           | 0.9538     | 0.9996     | 0.3773 | 0.9282 | 338.55           | 0.9828     | 0.9921     |
| 0.1095                            | 0.5758 | 370.47           | 0.9560     | 0.9993     | 0.4003 | 0.9371 | 336.44           | 0.9832     | 0.9914     |
| 0.1263                            | 0.6261 | 367.60           | 0.9581     | 0.9990     | 0.4559 | 0.9539 | 332.60           | 0.9864     | 0.9890     |
| 0.1425                            | 0.6684 | 365.82           | 0.9605     | 0.9987     | 0.4909 | 0.9620 | 330.07           | 0.9885     | 0.9874     |
| 0.1608                            | 0.7099 | 363.12           | 0.9627     | 0.9984     | 0.5267 | 0.9687 | 327.74           | 0.9903     | 0.9858     |
| 0.1709                            | 0.7302 | 361.33           | 0.9639     | 0.9982     | 0.5716 | 0.9754 | 324.95           | 0.9921     | 0.9836     |
| 0.1822                            | 0.7510 | 359.88           | 0.9650     | 0.9980     | 0.6252 | 0.9817 | 322.08           | 0.9942     | 0.9818     |
| 0.1938                            | 0.7705 | 357.22           | 0.9664     | 0.9977     | 0.6588 | 0.9848 | 320.22           | 0.9953     | 0.9789     |
| 0.2075                            | 0.7913 | 356.57           | 0.9677     | 0.9974     | 0.6995 | 0.9880 | 318.36           | 0.9963     | 0.9765     |
| 0.2228                            | 0.8119 | 354.64           | 0.9694     | 0.9969     | 0.7462 | 0.9910 | 316.89           | 0.9975     | 0.9740     |
| 0.2406                            | 0.8330 | 352.53           | 0.9713     | 0.9966     | 0.7963 | 0.9936 | 314.67           | 0.9984     | 0.9708     |
| 0.2617                            | 0.8546 | 349.41           | 0.9731     | 0.9959     | 0.8572 | 0.9961 | 312.85           | 0.9993     | 0.9668     |
| 0.2870                            | 0.8763 | 347.08           | 0.9753     | 0.9955     | 0.9248 | 0.9982 | 309.20           | 0.9998     | 0.9620     |
| 0.2854                            | 0.8750 | 346.36           | 0.9752     | 0.9953     | 1.0000 | 1.0000 | 307.71           | 1.0000     |            |
| methanol (1) + 1-butanol (2)      |        |                  |            |            |        |        |                  |            |            |
| 0.0000                            | 0.0000 | 390.86           |            | 1.0000     | 0.6195 | 0.9295 | 349.14           | 0.9864     | 0.9717     |
| 0.0371                            | 0.1734 | 386.64           | 0.9359     | 0.9999     | 0.6467 | 0.9376 | 347.52           | 0.9881     | 0.9688     |
| 0.0812                            | 0.3328 | 381.87           | 0.9400     | 0.9996     | 0.6752 | 0.9455 | 346.41           | 0.9898     | 0.9655     |
| 0.1379                            | 0.4856 | 377.02           | 0.9454     | 0.9989     | 0.6740 | 0.9452 | 346.27           | 0.9897     | 0.9656     |
| 0.2041                            | 0.6140 | 370.83           | 0.9517     | 0.9976     | 0.6976 | 0.9512 | 345.63           | 0.9911     | 0.9628     |
| 0.2382                            | 0.6654 | 367.95           | 0.9549     | 0.9966     | 0.7231 | 0.9572 | 345.01           | 0.9924     | 0.9596     |
| 0.2817                            | 0.7200 | 365.80           | 0.9589     | 0.9951     | 0.7512 | 0.9632 | 344.28           | 0.9938     | 0.9558     |
| 0.3476                            | 0.7851 | 360.83           | 0.9649     | 0.9923     | 0.7813 | 0.9692 | 343.88           | 0.9952     | 0.9516     |
| 0.3818                            | 0.8124 | 359.35           | 0.9679     | 0.9905     | 0.8137 | 0.9751 | 342.73           | 0.9964     | 0.9468     |
| 0.4142                            | 0.8351 | 357.80           | 0.9707     | 0.9886     | 0.8494 | 0.9809 | 341.78           | 0.9976     | 0.9412     |
| 0.4455                            | 0.8544 | 356.24           | 0.9734     | 0.9866     | 0.8491 | 0.9808 | 341.79           | 0.9976     | 0.9413     |
| 0.4725                            | 0.8693 | 354.42           | 0.9756     | 0.9847     | 0.8722 | 0.9843 | 341.01           | 0.9983     | 0.9375     |
| 0.4969                            | 0.8816 | 353.60           | 0.9775     | 0.9829     | 0.8960 | 0.9877 | 340.28           | 0.9989     | 0.9334     |
| 0.5212                            | 0.8928 | 352.28           | 0.9794     | 0.9809     | 0.9193 | 0.9907 | 339.65           | 0.9993     | 0.9293     |
| 0.5411                            | 0.9012 | 351.77           | 0.9809     | 0.9792     | 0.9439 | 0.9938 | 338.98           | 0.9997     | 0.9248     |
| 0.5463                            | 0.9033 | 351.77           | 0.9813     | 0.9788     | 0.9722 | 0.9970 | 338.29           | 0.9999     | 0.9194     |
| 0.5679                            | 0.9117 | 350.56           | 0.9828     | 0.9768     | 1.0000 | 1.0000 | 337.66           | 1.0000     |            |
| 0.5931                            | 0.9208 | 349.61           | 0.9846     | 0.9744     |        |        |                  |            |            |



**Figure 1.**  $T$ – $x_1$ – $y_1$  diagram for diethyl ether (1) + methanol (2) at 101.325 kPa:  $\circ$ , experimental data; —, Wilson correlation; ---, ASOG prediction.



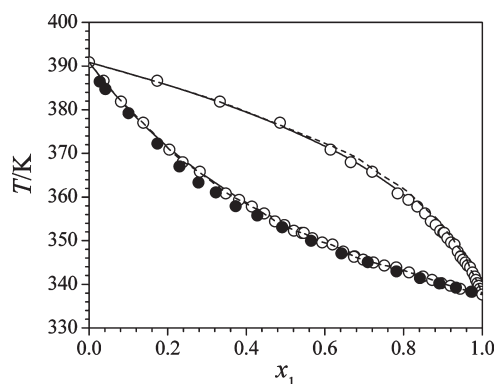
**Figure 2.**  $T$ – $x_1$ – $y_1$  diagram for diethyl ether (1) + 1-butanol (2) at 101.325 kPa:  $\circ$ , experimental data; —, Wilson correlation; ---, ASOG prediction.

the least-squares fitting as follows:

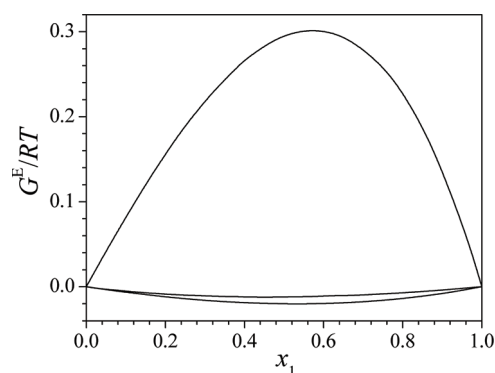
$$\text{OF} = \sum_{i=1}^N (x_{i,\text{calc}} - x_{i,\text{exp}})^2 \quad (2)$$

where  $x_{i,\text{calc}}$  and  $x_{i,\text{exp}}$  are the liquid-phase mole fraction of component  $i$  calculated and experimental values from eq 1 and from measured data, respectively.

VLE data for the three binary systems at 101.325 kPa including the measured liquid- and vapor-phase mole fraction  $x_1$  and  $y_1$ , experimental boiling point temperature  $T_{\text{exp}}$ , and experimental activity coefficients  $\gamma_1$  and  $\gamma_2$  are presented in Table 1. The  $T$ – $x_1$ – $y_1$  phase diagrams for these binary systems are shown in Figures 1–3. Also, the ASOG method was used to obtain the prediction for vapor-phase mole fractions, as shown in Figures 1–3. Moreover, the VLE data for the methanol (1) + 1-butanol (2) binary system compared with the previous literature<sup>14</sup> are illustrated in Figure 3. Comparing the results demonstrated that the values of liquid-phase mole fraction  $x_1$  well accorded with those of the literature. However, the VLE data for the other systems from these sources were not added into Figures 1–3 because they are either isothermal data or isobaric data (not at 101.325 kPa). The activity coefficients for three binary systems with Wilson model were used to evaluate dimensionless excess Gibbs energy function over the overall



**Figure 3.**  $T$ – $x_1$ – $y_1$  diagram for methanol (1) + 1-butanol (2) at 101.325 kPa:  $\circ$ , experimental data; —, Wilson correlation; ---, ASOG prediction;  $\bullet$ , from literature<sup>14</sup>.



**Figure 4.** Diagram for excess Gibbs energy functions ( $G^E/RT$ ) versus liquid-phase mole fraction of component 1 ( $x_1$ ). From top to bottom: diethyl ether (1) + methanol (2), diethyl ether (1) + 1-butanol (2), and methanol (1) + 1-butanol (2).

range of compositions at 101.325 kPa. The plot of excess Gibbs energy function  $G^E/RT$  versus liquid-phase mole fraction  $x_1$  for the three binary systems is given in Figure 4. All the mixtures exhibit deviations from ideality with a range that may be attributed to interactions leading to the formation of hydrogen bond between the polar groups. Observed nonideal behavior is indicative of the magnitude of experimental activity coefficients  $\gamma_i$  as well as of the variation of excess Gibbs energy function,  $G^E/RT$ , with composition, as depicted in Figure 4. The obtained absolute maximum values of  $G^E/RT$  for the diethyl ether + methanol (2), diethyl ether (1) + 1-butanol (2), and methanol (1) + 1-butanol (2) binary systems are 0.3019, 0.0123, and 0.0201, respectively. The values of excess Gibbs energy function  $G^E/RT$  are negative for the diethyl ether (1) + 1-butanol (2) and methanol (1) + 1-butanol (2) binary systems. However, for diethyl ether (1) + methanol (2) system, the values of those are positive in the overall range of mole fraction. The absolute maximum values of  $G^E/RT$  follow the order diethyl ether (1) + methanol (2) > methanol (1) + 1-butanol (2) > diethyl ether (1) + 1-butanol (2), and those of  $G^E/RT$  are approximate at an equimolar fraction in three binary systems. The optimum model interaction parameters of liquid-phase activity coefficient and the absolute average deviations ( $dT$ ,  $dy_1$ , and  $d\gamma$ ) are listed in Table 2. Herein, the obtained results revealed that the deviations of Wilson, NRTL, Margules, van Laar, and UNIQUAC equations

Table 2. Correlation Parameters for Activity Coefficients and Average Deviation for Studied Systems

| equation              | parameters or deviations                       | diethyl ether (1) + methanol (2) | diethyl ether (1) + 1-butanol (2) | methanol(1) + 1-butanol (2) |
|-----------------------|------------------------------------------------|----------------------------------|-----------------------------------|-----------------------------|
| Wilson <sup>a</sup>   | $\Lambda_{12}/\text{J}\cdot\text{mol}^{-1}$    | −516                             | −87                               | 12                          |
|                       | $\Lambda_{21}/\text{J}\cdot\text{mol}^{-1}$    | 1640                             | 65                                | 110                         |
|                       | $dT/\text{K}$                                  | 0.26                             | 0.40                              | 0.32                        |
|                       | $dy_1$                                         | 0.0102                           | 0.0096                            | 0.0072                      |
|                       | $d\gamma_1$                                    | 0.0319                           | 0.0160                            | 0.0033                      |
| NRTL <sup>a</sup>     | $d\gamma_2$                                    | 0.0667                           | 0.0078                            | 0.0030                      |
|                       | $(g_{12}-g_{11})/\text{J}\cdot\text{mol}^{-1}$ | −156                             | 210                               | −56.76                      |
|                       | $(g_{21}-g_{22})/\text{J}\cdot\text{mol}^{-1}$ | 710                              | −40                               | 0.48                        |
|                       | $\alpha_{12}$                                  | −0.40                            | 0.30                              | 0.36                        |
|                       | $dT/\text{K}$                                  | 0.28                             | 0.45                              | 0.26                        |
| Margules <sup>b</sup> | $dy_1$                                         | 0.0105                           | 0.0116                            | 0.0068                      |
|                       | $d\gamma_1$                                    | 0.0202                           | 0.0065                            | 0.0028                      |
|                       | $d\gamma_2$                                    | 0.0495                           | 0.0043                            | 0.0025                      |
|                       | $A_{12}$                                       | 0.75                             | −0.06                             | −0.08                       |
|                       | $A_{21}$                                       | 1.56                             | −0.12                             | −0.10                       |
| van Laar <sup>b</sup> | $dT/\text{K}$                                  | 0.32                             | 0.34                              | 0.31                        |
|                       | $dy_1$                                         | 0.0112                           | 0.0110                            | 0.0072                      |
|                       | $d\gamma_1$                                    | 0.0331                           | 0.0054                            | 0.0003                      |
|                       | $d\gamma_2$                                    | 0.0369                           | 0.0034                            | 0.0004                      |
|                       | $A_{12}$                                       | 0.78                             | −0.078                            | −0.08                       |
| UNIQUAC <sup>a</sup>  | $A_{21}$                                       | 1.60                             | −0.068                            | −0.10                       |
|                       | $dT/\text{K}$                                  | 0.27                             | 0.32                              | 0.28                        |
|                       | $dy_1$                                         | 0.0113                           | 0.0110                            | 0.0068                      |
|                       | $d\gamma_1$                                    | 0.0188                           | 0.0077                            | 0.0017                      |
|                       | $d\gamma_2$                                    | 0.0617                           | 0.0037                            | 0.0026                      |
| UNIQUAC <sup>a</sup>  | $u_{12}/\text{J}\cdot\text{mol}^{-1}$          | 1647.76                          | 1548.75                           | 1846.76                     |
|                       | $u_{21}/\text{J}\cdot\text{mol}^{-1}$          | 2688.35                          | 2886.68                           | 2986.42                     |
|                       | $dT/\text{K}$                                  | 0.34                             | 0.46                              | 0.31                        |
|                       | $dy_1$                                         | 0.0084                           | 0.0085                            | 0.0076                      |
|                       | $d\gamma_1$                                    | 0.0125                           | 0.0046                            | 0.0036                      |
|                       | $d\gamma_2$                                    | 0.0145                           | 0.0038                            | 0.0028                      |

<sup>a</sup> Wilson, NRTL, and UNIQUAC's interaction parameters,  $\text{J}\cdot\text{mol}^{-1}$ . <sup>b</sup> Margules and van Laar's interaction parameters, dimensionless.  $dT = \Sigma |T_{\text{exp}} - T_{\text{calc}}|/N$ .  $T_{\text{exp}}$ : experimental boiling point temperature, K.  $T_{\text{calc}}$ : calculated bubble point from model, K.  $dy = \Sigma |y_{\text{exp}} - y_{\text{calc}}|/N$ .  $y_{\text{exp}}$ : experimental vapor-phase mole fraction,  $y_{\text{calc}}$ : calculated vapor-phase mole fraction from model,  $d\gamma_i = \Sigma |\gamma_{i,\text{exp}} - \gamma_{i,\text{calc}}|/N$ .  $\gamma_{i,\text{exp}}$ : experimental values of component  $i$ .  $\gamma_{i,\text{calc}}$ : calculated values of component  $i$  from model.  $N$ : number of data points.

were reasonably small as shown in Table 2. Because the superiority of one method over the others is not always obvious, practice must rely on experience and analogy. Comprehensive comparisons of five of the methods (Wilson, NRTL, Margules, van Laar, and UNIQUAC) are made in Table 2. From the data analysis, the temperature deviations between the experimental and calculated values of five different types of model are very similar in the three binary systems, and the vapor-phase mole fraction deviations between the values from the measurement and from the model are very similar. Therefore, the activity coefficient models are appropriate for representing the experimental data of the three binary systems. In Table 2, the absolute average deviations  $dT$  between boiling point temperature from experiment and bubbling point temperature from calculation with Wilson model are 0.26 K, 0.40 K, and 0.32 K, respectively. Moreover, the absolute average deviations  $dy_1$  between vapor-phase mole fraction from experiment and from calculation with Wilson model are 0.0102, 0.0096, and 0.0072, respectively. In addition, the average absolute deviations  $d\gamma_1$  and  $d\gamma_2$  between the values from experiment and from calculation with Wilson

model are 0.0319, 0.0667, 0.0160, 0.0078, and 0.0033, 0.0030, respectively. The results have demonstrated that the activity coefficients methods correlate well the experimental data. In Table 1, the experimental values of the liquid-phase activity coefficients  $\gamma_1$  and  $\gamma_2$  for the diethyl ether (1) + 1-butanol (2) and methanol (1) + 1-butanol (2) are less than 1, therefore, these binary systems show negative deviation from ideal behaviors. However, the diethyl ether (1) + methanol (2) binary system shows positive deviation from ideal behaviors because the values of those are more than 1.

#### Measurement and Prediction of VLE of Ternary System.

VLE data for the ternary system at 101.325 kPa including the measured liquid- and vapor-phase mole fraction  $x_1$ ,  $x_2$ , and  $x_3$ ,  $y_1$ ,  $y_2$ , and  $y_3$ , experimental boiling point temperature  $T_{\text{exp}}$ , experimental activity coefficients  $\gamma_1$ ,  $\gamma_2$ , and  $\gamma_3$ , are listed in Table 3. In addition, the binary interaction parameters of the Wilson, NRTL, Margules, van Laar, and UNIQUAC equation presented in Table 2 were used to predict the VLE data for the ternary system, which obtained the bubble points, calculated vapor-phase compositions, and activity coefficients compared

**Table 3. VLE Data for the Diethyl Ether (1) + Methanol (2) + 1-Butanol Ternary System at 101.325 kPa: Measured Liquid-Phase Mole Fraction  $x_1$ ,  $x_2$ , and  $x_3$ , Experimental Boiling Point Temperature  $T_{\text{exp}}$ , Measured Vapor-Phase Mole Fraction  $y_1$ ,  $y_2$ , and  $y_3$ , and Experimental Activity Coefficients  $\gamma_1$ ,  $\gamma_2$ , and  $\gamma_3$**

| run no. | $x_1$  | $x_2$  | $x_3$  | $T_{\text{exp}} / \text{K}$ | $y_1$  | $y_2$  | $y_3$  | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ |
|---------|--------|--------|--------|-----------------------------|--------|--------|--------|------------|------------|------------|
| 1       | 0.8654 | 0.0980 | 0.0366 | 308.50                      | 0.9184 | 0.0811 | 0.0005 | 1.0429     | 2.7723     | 0.6162     |
| 2       | 0.8416 | 0.0954 | 0.0630 | 309.15                      | 0.9226 | 0.0764 | 0.0010 | 1.0467     | 2.5766     | 0.6665     |
| 3       | 0.8230 | 0.0932 | 0.0838 | 309.59                      | 0.9254 | 0.0732 | 0.0014 | 1.0483     | 2.4423     | 0.7005     |
| 4       | 0.8036 | 0.0911 | 0.1053 | 310.04                      | 0.9278 | 0.0703 | 0.0020 | 1.0493     | 2.3169     | 0.7311     |
| 5       | 0.7819 | 0.0886 | 0.1295 | 310.63                      | 0.9300 | 0.0674 | 0.0027 | 1.0496     | 2.1925     | 0.7612     |
| 6       | 0.7633 | 0.0865 | 0.1502 | 311.15                      | 0.9314 | 0.0652 | 0.0033 | 1.0493     | 2.0971     | 0.7838     |
| 7       | 0.7378 | 0.0836 | 0.1786 | 312.86                      | 0.9330 | 0.0626 | 0.0044 | 1.0482     | 1.9808     | 0.8109     |
| 8       | 0.7150 | 0.0810 | 0.2040 | 313.95                      | 0.9339 | 0.0606 | 0.0055 | 1.0468     | 1.8889     | 0.8319     |
| 9       | 0.6928 | 0.0785 | 0.2287 | 314.58                      | 0.9343 | 0.0590 | 0.0067 | 1.0451     | 1.8087     | 0.8499     |
| 10      | 0.6728 | 0.0762 | 0.2510 | 315.15                      | 0.9345 | 0.0576 | 0.0079 | 1.0432     | 1.7432     | 0.8644     |
| 11      | 0.6519 | 0.0739 | 0.2742 | 316.80                      | 0.9343 | 0.0564 | 0.0093 | 1.0412     | 1.6808     | 0.8779     |
| 12      | 0.6384 | 0.0723 | 0.2893 | 317.25                      | 0.9340 | 0.0556 | 0.0104 | 1.0397     | 1.6434     | 0.8860     |
| 13      | 0.5084 | 0.2549 | 0.2367 | 316.80                      | 0.8327 | 0.1603 | 0.0070 | 1.1769     | 1.3778     | 0.7542     |
| 14      | 0.4968 | 0.2491 | 0.2541 | 317.54                      | 0.8312 | 0.1607 | 0.0081 | 1.1707     | 1.3621     | 0.7715     |
| 15      | 0.4831 | 0.2423 | 0.2746 | 319.35                      | 0.8293 | 0.1613 | 0.0095 | 1.1634     | 1.3437     | 0.7906     |
| 16      | 0.4738 | 0.2376 | 0.2886 | 320.16                      | 0.8279 | 0.1616 | 0.0106 | 1.1583     | 1.3315     | 0.8029     |
| 17      | 0.4634 | 0.2324 | 0.3042 | 320.86                      | 0.8262 | 0.1620 | 0.0118 | 1.1527     | 1.3179     | 0.8159     |
| 18      | 0.4527 | 0.2270 | 0.3203 | 321.53                      | 0.8243 | 0.1624 | 0.0133 | 1.1468     | 1.3041     | 0.8285     |
| 19      | 0.4410 | 0.2211 | 0.3379 | 322.98                      | 0.8221 | 0.1629 | 0.0150 | 1.1404     | 1.2894     | 0.8415     |
| 20      | 0.4304 | 0.2158 | 0.3538 | 323.35                      | 0.8199 | 0.1634 | 0.0168 | 1.1346     | 1.2762     | 0.8526     |
| 21      | 0.4196 | 0.2104 | 0.3700 | 324.04                      | 0.8174 | 0.1639 | 0.0187 | 1.1287     | 1.2631     | 0.8632     |
| 22      | 0.4106 | 0.2059 | 0.3835 | 324.74                      | 0.8153 | 0.1643 | 0.0204 | 1.1238     | 1.2524     | 0.8716     |
| 23      | 0.4026 | 0.2019 | 0.3955 | 325.45                      | 0.8132 | 0.1647 | 0.0221 | 1.1194     | 1.2430     | 0.8787     |
| 24      | 0.3928 | 0.1970 | 0.4102 | 327.06                      | 0.8106 | 0.1652 | 0.0243 | 1.1141     | 1.2316     | 0.8869     |
| 25      | 0.3845 | 0.1928 | 0.4227 | 328.00                      | 0.8082 | 0.1655 | 0.0263 | 1.1096     | 1.2222     | 0.8936     |
| 26      | 0.3751 | 0.1881 | 0.4368 | 328.04                      | 0.8053 | 0.1660 | 0.0287 | 1.1045     | 1.2117     | 0.9007     |
| 27      | 0.3677 | 0.1844 | 0.4479 | 328.75                      | 0.8029 | 0.1664 | 0.0308 | 1.1005     | 1.2036     | 0.9061     |
| 28      | 0.0767 | 0.9233 | 0.0000 | 327.87                      | 0.3493 | 0.6507 | 0.0000 | 2.3388     | 1.0040     | 0.7830     |
| 29      | 0.0753 | 0.9069 | 0.0178 | 328.96                      | 0.3390 | 0.6600 | 0.0010 | 2.2633     | 1.0066     | 0.7943     |
| 30      | 0.0741 | 0.8925 | 0.0334 | 329.24                      | 0.3304 | 0.6676 | 0.0020 | 2.2012     | 1.0087     | 0.8037     |
| 31      | 0.0728 | 0.8764 | 0.0508 | 329.98                      | 0.3213 | 0.6754 | 0.0032 | 2.1361     | 1.0106     | 0.8138     |
| 32      | 0.0716 | 0.8619 | 0.0665 | 330.97                      | 0.3134 | 0.6822 | 0.0044 | 2.0808     | 1.0121     | 0.8226     |
| 33      | 0.0705 | 0.8482 | 0.0813 | 330.96                      | 0.3063 | 0.6880 | 0.0056 | 2.0314     | 1.0133     | 0.8306     |
| 34      | 0.0693 | 0.8345 | 0.0962 | 331.87                      | 0.2992 | 0.6939 | 0.0069 | 1.9845     | 1.0143     | 0.8384     |
| 35      | 0.0681 | 0.8200 | 0.1119 | 332.54                      | 0.2921 | 0.6995 | 0.0084 | 1.9374     | 1.0152     | 0.8463     |
| 36      | 0.0669 | 0.8059 | 0.1272 | 332.86                      | 0.2853 | 0.7047 | 0.0100 | 1.8939     | 1.0158     | 0.8538     |
| 37      | 0.0658 | 0.7921 | 0.1421 | 333.86                      | 0.2792 | 0.7093 | 0.0116 | 1.8535     | 1.0163     | 0.8606     |
| 38      | 0.0216 | 0.9784 | 0.0000 | 334.00                      | 0.1270 | 0.8730 | 0.0000 | 2.5224     | 1.0003     | 0.8805     |
| 39      | 0.0645 | 0.7767 | 0.1588 | 334.10                      | 0.2723 | 0.7142 | 0.0135 | 1.8105     | 1.0166     | 0.8681     |
| 40      | 0.0213 | 0.9657 | 0.0130 | 334.73                      | 0.1231 | 0.8758 | 0.0011 | 2.4530     | 1.0009     | 0.8839     |
| 41      | 0.0633 | 0.7626 | 0.1741 | 334.85                      | 0.2662 | 0.7184 | 0.0154 | 1.7732     | 1.0167     | 0.8747     |
| 42      | 0.0210 | 0.9522 | 0.0268 | 335.41                      | 0.1193 | 0.8783 | 0.0024 | 2.3835     | 1.0015     | 0.8875     |
| 43      | 0.0622 | 0.7492 | 0.1886 | 335.44                      | 0.2607 | 0.7220 | 0.0173 | 1.7393     | 1.0167     | 0.8807     |
| 44      | 0.0611 | 0.7359 | 0.2030 | 336.14                      | 0.2554 | 0.7253 | 0.0193 | 1.7072     | 1.0165     | 0.8864     |
| 45      | 0.0207 | 0.9382 | 0.0411 | 336.22                      | 0.1156 | 0.8806 | 0.0038 | 2.3158     | 1.0020     | 0.8912     |
| 46      | 0.0204 | 0.9239 | 0.0557 | 336.64                      | 0.1121 | 0.8827 | 0.0052 | 2.2508     | 1.0024     | 0.8949     |
| 47      | 0.0148 | 0.9575 | 0.0277 | 336.93                      | 0.0866 | 0.9108 | 0.0026 | 2.3985     | 1.0010     | 0.8989     |
| 48      | 0.0599 | 0.7209 | 0.2192 | 336.94                      | 0.2497 | 0.7286 | 0.0217 | 1.6726     | 1.0162     | 0.8926     |
| 49      | 0.0201 | 0.9101 | 0.0698 | 337.15                      | 0.1088 | 0.8845 | 0.0067 | 2.1917     | 1.0027     | 0.8984     |
| 50      | 0.0588 | 0.7079 | 0.2333 | 337.45                      | 0.2448 | 0.7314 | 0.0239 | 1.6439     | 1.0158     | 0.8978     |
| 51      | 0.0144 | 0.9330 | 0.0526 | 337.55                      | 0.0817 | 0.9131 | 0.0052 | 2.2810     | 1.0015     | 0.9042     |
| 52      | 0.0197 | 0.8923 | 0.0880 | 337.65                      | 0.1047 | 0.8865 | 0.0088 | 2.1204     | 1.0030     | 0.9029     |

Table 3. Continued

| run no. | $x_1$  | $x_2$  | $x_3$  | $T_{\text{exp}} / \text{K}$ | $y_1$  | $y_2$  | $y_3$  | $\gamma_1$ | $\gamma_2$ | $\gamma_3$ |
|---------|--------|--------|--------|-----------------------------|--------|--------|--------|------------|------------|------------|
| 53      | 0.0579 | 0.6965 | 0.2456 | 337.86                      | 0.2407 | 0.7334 | 0.0259 | 1.6197     | 1.0154     | 0.9021     |
| 54      | 0.014  | 0.9066 | 0.0794 | 338.15                      | 0.0771 | 0.9148 | 0.0081 | 2.1677     | 1.0019     | 0.9098     |
| 55      | 0.0193 | 0.8760 | 0.1047 | 338.17                      | 0.1011 | 0.8882 | 0.0108 | 2.0594     | 1.0031     | 0.9070     |
| 56      | 0.0190 | 0.8607 | 0.1203 | 338.76                      | 0.0981 | 0.8891 | 0.0127 | 2.0057     | 1.0032     | 0.9106     |
| 57      | 0.0136 | 0.8780 | 0.1084 | 338.94                      | 0.0727 | 0.9156 | 0.0117 | 2.0582     | 1.0020     | 0.9157     |
| 58      | 0.0186 | 0.8433 | 0.1381 | 339.45                      | 0.0947 | 0.8902 | 0.0151 | 1.9483     | 1.0032     | 0.9147     |
| 59      | 0.0132 | 0.8528 | 0.1340 | 339.79                      | 0.0690 | 0.9159 | 0.0151 | 1.9715     | 1.0018     | 0.9209     |
| 60      | 0.0182 | 0.8270 | 0.1548 | 339.88                      | 0.0916 | 0.8910 | 0.0174 | 1.8978     | 1.0030     | 0.9186     |
| 61      | 0.0128 | 0.8302 | 0.1570 | 340.53                      | 0.0657 | 0.9159 | 0.0184 | 1.9004     | 1.0015     | 0.9255     |
| 62      | 0.0125 | 0.8087 | 0.1788 | 341.08                      | 0.0632 | 0.9151 | 0.0218 | 1.8382     | 1.0011     | 0.9296     |
| 63      | 0.0122 | 0.7870 | 0.2008 | 341.55                      | 0.0608 | 0.9138 | 0.0254 | 1.7801     | 1.0006     | 0.9337     |
| 64      | 0.0119 | 0.7700 | 0.2181 | 342.26                      | 0.0588 | 0.9128 | 0.0284 | 1.7374     | 1.0000     | 0.9369     |
| 65      | 0.0264 | 0.6384 | 0.3352 | 344.49                      | 0.1187 | 0.8316 | 0.0498 | 1.4884     | 1.0005     | 0.9454     |
| 66      | 0.0259 | 0.6270 | 0.3471 | 344.98                      | 0.1164 | 0.8308 | 0.0529 | 1.4696     | 0.9996     | 0.9477     |
| 67      | 0.0252 | 0.6087 | 0.3661 | 344.96                      | 0.1132 | 0.8287 | 0.0580 | 1.4407     | 0.9983     | 0.9513     |
| 68      | 0.0245 | 0.5935 | 0.3820 | 345.98                      | 0.1102 | 0.8271 | 0.0627 | 1.4176     | 0.9970     | 0.9542     |
| 69      | 0.0239 | 0.5786 | 0.3975 | 346.42                      | 0.1076 | 0.8249 | 0.0675 | 1.3960     | 0.9957     | 0.9569     |
| 70      | 0.0234 | 0.5655 | 0.4111 | 347.07                      | 0.1055 | 0.8226 | 0.0719 | 1.3777     | 0.9946     | 0.9592     |
| 71      | 0.0229 | 0.5546 | 0.4225 | 347.55                      | 0.1034 | 0.8208 | 0.0758 | 1.3629     | 0.9936     | 0.9611     |
| 72      | 0.0224 | 0.5416 | 0.4360 | 348.09                      | 0.1014 | 0.8180 | 0.0806 | 1.3459     | 0.9924     | 0.9632     |
| 73      | 0.0408 | 0.4577 | 0.5015 | 350.47                      | 0.1774 | 0.7243 | 0.0983 | 1.2585     | 0.9947     | 0.9650     |
| 74      | 0.0398 | 0.4464 | 0.5138 | 351.07                      | 0.1741 | 0.7219 | 0.1039 | 1.2462     | 0.9932     | 0.9671     |
| 75      | 0.0386 | 0.4332 | 0.5282 | 351.85                      | 0.1702 | 0.7189 | 0.1109 | 1.2321     | 0.9914     | 0.9694     |
| 76      | 0.0372 | 0.4171 | 0.5457 | 352.66                      | 0.1657 | 0.7143 | 0.1200 | 1.2156     | 0.9892     | 0.9720     |
| 77      | 0.0360 | 0.4041 | 0.5599 | 353.55                      | 0.1617 | 0.7104 | 0.1279 | 1.2026     | 0.9874     | 0.9741     |
| 78      | 0.0345 | 0.3875 | 0.5780 | 354.35                      | 0.1568 | 0.7046 | 0.1386 | 1.1866     | 0.9850     | 0.9765     |
| 79      | 0.0329 | 0.3691 | 0.5980 | 355.68                      | 0.1516 | 0.6971 | 0.1514 | 1.1695     | 0.9823     | 0.9790     |
| 80      | 0.0311 | 0.3489 | 0.6200 | 357.25                      | 0.1456 | 0.6876 | 0.1668 | 1.1515     | 0.9793     | 0.9816     |
| 81      | 0.0300 | 0.3371 | 0.6329 | 357.98                      | 0.1418 | 0.6816 | 0.1765 | 1.1413     | 0.9775     | 0.9830     |
| 82      | 0.0326 | 0.2410 | 0.7264 | 362.65                      | 0.1636 | 0.5793 | 0.2571 | 1.0696     | 0.9682     | 0.9900     |
| 83      | 0.0315 | 0.2326 | 0.7359 | 364.36                      | 0.1598 | 0.5717 | 0.2685 | 1.0635     | 0.9667     | 0.9907     |
| 84      | 0.0303 | 0.2238 | 0.7459 | 364.32                      | 0.1555 | 0.5633 | 0.2812 | 1.0572     | 0.9650     | 0.9915     |
| 85      | 0.0292 | 0.2159 | 0.7549 | 365.22                      | 0.1515 | 0.5553 | 0.2933 | 1.0515     | 0.9635     | 0.9921     |
| 86      | 0.0281 | 0.2077 | 0.7642 | 366.44                      | 0.1474 | 0.5464 | 0.3062 | 1.0458     | 0.9620     | 0.9928     |
| 87      | 0.0272 | 0.2008 | 0.7720 | 368.00                      | 0.1441 | 0.5384 | 0.3175 | 1.0410     | 0.9608     | 0.9933     |
| 88      | 0.0262 | 0.1933 | 0.7805 | 368.76                      | 0.1403 | 0.5293 | 0.3304 | 1.0359     | 0.9594     | 0.9938     |
| 89      | 0.0254 | 0.1880 | 0.7866 | 369.21                      | 0.1371 | 0.5228 | 0.3401 | 1.0323     | 0.9583     | 0.9942     |
| 90      | 0.0248 | 0.1834 | 0.7918 | 368.44                      | 0.1347 | 0.5168 | 0.3485 | 1.0292     | 0.9575     | 0.9945     |

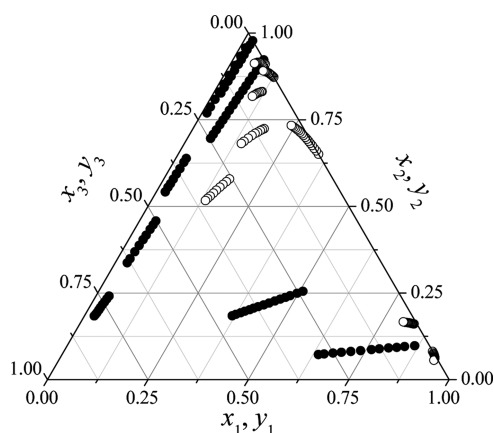
Table 4. Average Absolute Deviations for the Ternary System for Equilibrium Temperature, Vapor-Phase Composition, and Liquid-Phase Activity Coefficient

| model    | dT   | $dy_1$ | $dy_2$ | $dy_3$ | $d\gamma_1$ | $d\gamma_2$ | $d\gamma_3$ |
|----------|------|--------|--------|--------|-------------|-------------|-------------|
| Wilson   | 0.55 | 0.0168 | 0.0136 | 0.0204 | 0.0218      | 0.0324      | 0.0216      |
| NRTL     | 0.53 | 0.0138 | 0.0117 | 0.0213 | 0.0210      | 0.0317      | 0.0213      |
| Margules | 0.63 | 0.0145 | 0.0138 | 0.0232 | 0.0225      | 0.0338      | 0.0241      |
| van Laar | 0.66 | 0.0142 | 0.0146 | 0.0245 | 0.0237      | 0.0352      | 0.0240      |
| UNIQUAC  | 0.78 | 0.0171 | 0.0197 | 0.0262 | 0.0265      | 0.0381      | 0.0255      |

with the measured values. The average absolute deviations for equilibrium temperature, vapor-phase composition, and liquid activity coefficient are shown in Table 4. The absolute average and maximum deviation between the boiling point from experimental data and the bubble point from Wilson model

calculation are 0.54 and 1.37 K, respectively. Meanwhile, the average and maximum deviations using NRTL, Margules, van Laar, and UNIQUAC equation individually are 0.53 K, 1.38 K, 0.63 K, 1.40 K, 0.66 K, 1.42 and 0.78 K, 1.50 K. The absolute average deviations for vapor-phase composition using Wilson model are 0.0168, 0.0136, and 0.0204, respectively. The absolute average deviations for liquid-phase activity coefficient using Wilson model are 0.0218, 0.0324, and 0.0216, as shown in Table 4. From these data, the results with NRTL are relatively better than those of the other equations. Diagram of VLE for the ternary system diethyl ether (1) + methanol (2) + 1-butanol (3) at 101.325 kPa is shown in Figure 5.

**Thermodynamic Consistency Tests Based on VLE Data.** Consistency tests are techniques that allow, in principle, the assessment of experimental VLE data on the basis of the Gibbs–Duhem equation.<sup>33</sup> Herein, the thermodynamic consistency of



**Figure 5.** Diagram of VLE for the ternary system diethyl ether (1) + methanol (2) + 1-butanol (3) at 101.325 kPa: ●, liquid-phase mole fraction; ○, vapor-phase mole fraction.

the experimental VLE data was checked by means of the Wisniak's  $L-W$  test for these binary systems,<sup>26</sup> and the Wisniak–Tamir's modification of McDermott–Ellis test for the ternary system.<sup>27,28</sup> For these binary systems, if the VLE data are thermodynamically consistent, the values of  $L_i$  and  $W_i$  should be approximately identical. The ratios of  $L_i$  to  $W_i$  for the three binary systems all approach the value of 1 ( $0.92 < L_i/W_i < 1.08$  at all data points). That is to say, the binary data passed Wisniak's  $L-W$  test of the thermodynamic consistency (a value of  $D < 3$  confirms overall consistency). For the ternary system, in the modified McDermott–Ellis test, local deviations ( $D$ ) for the system diethyl ether + methanol + 1-butanol did not exceed 0.00720, while the maximum deviation was 0.06858. Thus, for this ternary system,  $D$  less than  $D_{\max}$  for all points in the modified McDermott–Ellis test confirms the thermodynamic consistency of the experimental VLE data.

## CONCLUSIONS

VLE data for the ternary system diethyl ether + methanol + 1-butanol and three constituent binary systems diethyl ether + methanol, diethyl ether + 1-butanol, and methanol + 1-butanol were determined by different liquid-phase compositions using a dynamic recirculating still at 101.325 kPa. The experimental data were correlated using the Wilson, NRTL, Margules, van Laar, and UNIQUAC equations, and the ASOG<sup>25</sup> model was used for prediction of the three binary systems. It was shown that the deviations of Wilson, NRTL, Margules, van Laar, and UNIQUAC equations are reasonably small. Moreover, the experimental results by comparison with the three binary systems of the correlation of the five models and prediction of the ASOG model are very similar. In addition, the VLE data of ternary system were predicted by the binary interaction parameters of Wilson, NRTL, Margules, van Laar, and UNIQUAC equations without any additional adjustment. The calculated bubble points accorded well with the experimental data. The results show that the calculated bubble point is fitted by the models which satisfy the need for the design and operation of separation process in chemistry industry. Moreover, the method will provide theoretical guidance for the research of VLE data of strongly associating system of vapor and liquid phase in nonideal behavior, and may be the indicator for the correlation and prediction of the methanol system VLE data.

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